

Variance ratio

$T = 10$

1-period

r_t

$t=1$	r_1
$t=2$	r_2
$t=3$	r_3
$t=4$	r_4
$\vdots$	$\vdots$
$t=10$	r_{10}
	↓ variance ↓ s_1^2

$n = 3$ -period

$$r_{3,t} = \frac{r_1 + r_2 + r_3}{r_t + r_{t-1} + r_{t-2}}$$

$t=1$	
$t=2$	
$t=3$	$r_{3,3} = r_3 + r_2 + r_1$
$t=4$	$r_{3,4} = r_4 + r_3 + r_2$
$\vdots$	$\vdots$
$t=10$	$r_{3,10} = r_{10} + r_9 + r_8$
	↓ variance ↓ $s_n^2 = s_3^2$

$$VR_n = VR_3 = \frac{s_3^2}{n s_1^2}$$

An alternative way to examine the efficient markets hypothesis is to compare the variance on returns over different time horizons.

Recall the 1-period log returns

Week 1 $r_t = \log P_t - \log P_{t-1}$

and the n -period returns

Week 2 $r_{nt} = \log P_t - \log P_{t-n} = \underline{r_t} + \underline{r_{t-1}} + \dots + r_{t-(n-1)},$

$n=2$

2-period { $E(r_t + r_{t-1}) = E(r_t) + E(r_{t-1}) = \mu + \mu = 2\mu$

$\text{var}(r_t + r_{t-1}) = \text{var}(r_t) + \text{var}(r_{t-1}) + 2 \underbrace{\text{cov}(r_t, r_{t-1})}_{=0 \text{ EMH}}$

$\quad \quad \quad = \sigma^2 + \sigma^2$

$\quad \quad \quad = 2\sigma^2$

1-period $\leftarrow \text{var}(r_t) = \sigma^2$

$S_2^2 \leftarrow$

2-period $\text{VR} = \frac{S_2^2}{n S_1^2} = \frac{2\sigma^2}{2\sigma^2} = 1$

1-period. $\leftarrow$ EMH holds.

Consider the variances of the 1-period returns and the n -period returns

$$s_1^2 = \frac{1}{T} \sum_{t=1}^T (r_t - \bar{r})^2, \quad s_n^2 = \frac{1}{T} \sum_{t=1}^T (r_{nt} - n\bar{r})^2, \quad \text{slide 58}$$

↓
slide 59

where $n\bar{r}$ represents the sample mean of n -period returns r_{nt} .

If there is no autocorrelation, the variance of n -period returns should equal n times the variance of the 1-period returns.

The ratio

$$VR_n = \frac{s_n^2}{ns_1^2},$$

is known as the variance ratio.

Variance ratio

The variance ratio has the following implications for the properties of log returns:

$$VR_n = \begin{cases} = 1 & \text{[No autocorrelation]} \Rightarrow \text{EMH holds} \\ > 1 & \text{[Positive autocorrelation]} \\ < 1 & \text{[Negative autocorrelation]} \end{cases} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{fails EMH}$$

Consider an $n = 3$ period return

$$r_{3t} = r_t + r_{t-1} + r_{t-2},$$

which is the sum of the three 1-period returns.

Let the sample mean for the 1-period returns be $\bar{r}$.

Subtracting $\bar{r}$ from both sides **three times** gives

$$(r_{3t} - 3\bar{r}) = (r_t - \bar{r}) + (r_{t-1} - \bar{r}) + (r_{t-2} - \bar{r}).$$

Cross product

By **squaring both sides** and averaging over a sample of size T , we obtain the expression shown in the next slide.

$$\begin{aligned}
& \frac{1}{T} \sum_{t=1}^T (r_{3t} - 3\bar{r})^2 \\
&= \underbrace{\frac{1}{T} \sum_{t=1}^T (r_t - \bar{r})^2}_{S_1^2} + \underbrace{\frac{1}{T} \sum_{t=1}^T (r_{t-1} - \bar{r})^2}_{S_1^2} + \underbrace{\frac{1}{T} \sum_{t=1}^T (r_{t-2} - \bar{r})^2}_{S_1^2} [\text{Var of } \underline{r_t}, \underline{r_{t-1}}, \underline{r_{t-2}}]
\end{aligned}$$

$$\begin{aligned}
& + \frac{2}{T} \sum_{t=1}^T (\underline{r_t} - \bar{r})(\underline{r_{t-1}} - \bar{r}) \quad [\text{Autocovariance of } r_t, r_{t-1}] \\
& + \frac{2}{T} \sum_{t=1}^T (\underline{r_t} - \bar{r})(\underline{r_{t-2}} - \bar{r}) \quad [\text{Autocovariance of } r_t, r_{t-2}] \\
& + \frac{2}{T} \sum_{t=1}^T (\underline{r_{t-1}} - \bar{r})(\underline{r_{t-2}} - \bar{r}) \quad [\text{Autocovariance of } r_{t-1}, r_{t-2}]
\end{aligned}$$

cross product

if
= 0
EMH
hold

Variance ratio

This expansion requires values for r_0 and r_{-1} . To implement this formulation in practice, the summation ranges are suitably adjusted to the available data.

In the case of zero autocovariances (no autocorrelation), the relationship simplifies to

$$s_3^2 = \underbrace{\frac{1}{T} \sum_{t=1}^T (r_t - \bar{r})^2}_{s_1^2} + \underbrace{\frac{1}{T} \sum_{t=1}^T (r_{t-1} - \bar{r})^2}_{s_1^2} + \underbrace{\frac{1}{T} \sum_{t=1}^T (r_{t-2} - \bar{r})^2}_{s_1^2}. \quad \text{stationarity}$$

$$s_3^2 = 3s_1^2$$

$$VR_3 = \frac{s_3^2}{n \cdot s_1^2} = \frac{3s_1^2}{3s_1^2} = 1$$

Variance ratio

Assuming that the variance for r_t is the same as the variance for r_{t-1} and r_{t-2} , then in the case of no autocorrelation in $n = 3$ period returns

$$s_3^2 = 3s_1^2.$$

The assumption of equal variance for r_{t-1} and r_{t-2} is known as stationarity.

Stationarity

The above measures can only be computed in a meaningful way if the time series is **stationary**.

Definition

A time series y_t is **(weakly or covariance) stationary** if for all t

1. $\mathbb{E}[y_t] = \mu$;
2. $\mathbb{V}[y_t] = \sigma^2$;
3. $\gamma(j)$ depends only on j but not on t , for $j = 1, 2, \dots$

Slide 58 $n = 10$

Suppose `r.xts` is your `xts` return series
 $r_t = r \leftarrow \text{as.numeric}(\text{na.omit}(r.xts)) \Rightarrow \text{return}$
`n <- length(r)`

Overlapping 3-period returns
 $r_{3,10} \Rightarrow r3 \leftarrow r[1:(n-2)] + r[2:(n-1)] + r[3:n]$
 $\quad \quad \quad 1:8 \quad \quad \quad 2:9 \quad \quad \quad 3:10$

Simple variance ratio for $k = 3$

`VR3 <- var(r3) / (3 * var(r))`

VR3

$\downarrow$
 $k=3$

Using returns on \$ / £

```
> r <- logret  
  
> n <- length(r)  
  
> # overlapping 3-period returns  
> r3 <- r[1:(n-2)] + r[2:(n-1)] + r[3:n]  
  
> # simple variance ratio for k = 3  
> VR3 <- var(r3) / (3 * var(r))  
  
> VR3  
bp  
bp 3.000257
```

$$VR_3 = 3$$

EMH fails.

Week 7: Modeling Predictable Returns

Motivation

1. The goal of investing is to earn a profit.
2. The most meaningful measure of profit for investors is returns.
3. As we've seen, there is **mixed evidence for the EMH**, i.e., seems to be satisfied for some stocks but not others $\Rightarrow$ validity of the EMH is questionable.
4. When EMH is not satisfied, then we can try to predict, or forecast, returns.

EMH fails $\Rightarrow$ Fama is wrong
 $\Rightarrow$ predict r_t using $\overbrace{r_{t-1}, r_{t-2}, \dots}^{\text{past returns}}$
 $\underbrace{\hspace{10em}}_{\text{estimate AR(1) model}}$

Returns, Prices, without the EMH

- ▶ When there is evidence to suggest EMH is not satisfied...
- ▶ We want to model the dynamics of returns and use them to generate forecast for returns and for prices.
- ▶ The most common (and arguably the simplest) approaches to forecast are autoregressive (AR) models.
- ▶ First, we need to *verify* that returns are stationary.

Revision of properties of expectations

There are some simple properties of expectations. If y is a random variable and a is a constant then

$$E(a) = a \quad \text{var}(a) = 0$$


$$E(ay) = aE(y)$$


$$\text{var}(ay) = a^2 \text{var}(y)$$


$$E(a + y) = a + E(y)$$


$$\text{var}(a + y) = \text{var}(y)$$

Revision of properties of expectations

If y_1 and y_2 are random variables then

$$E(y_1 + y_2) = E(y_1) + E(y_2)$$


and

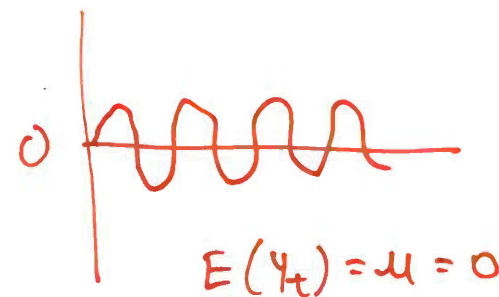
$$\text{var}(y_1 + y_2) = \text{var}(y_1) + \text{var}(y_2) + \underline{2 \text{cov}(y_1, y_2)}.$$


Stationary

$$Y_t \Rightarrow \text{R.V.}$$

$$Y_{t-1} \Rightarrow \text{R.V.}$$

A time series Y_t is defined to be covariance *stationary* if



1. $E(Y_t) = \mu$ does not depend on t

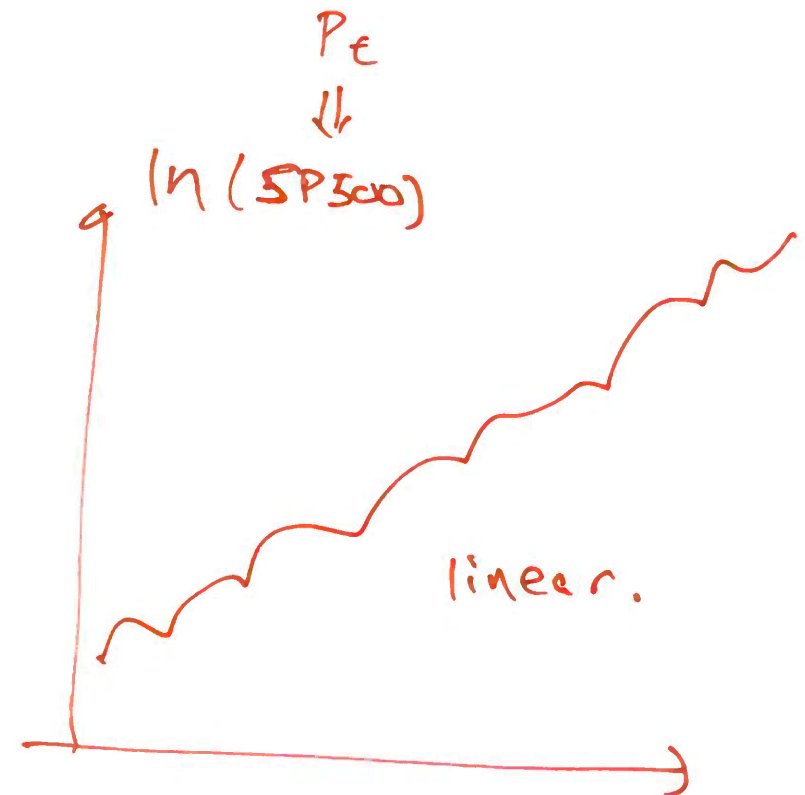
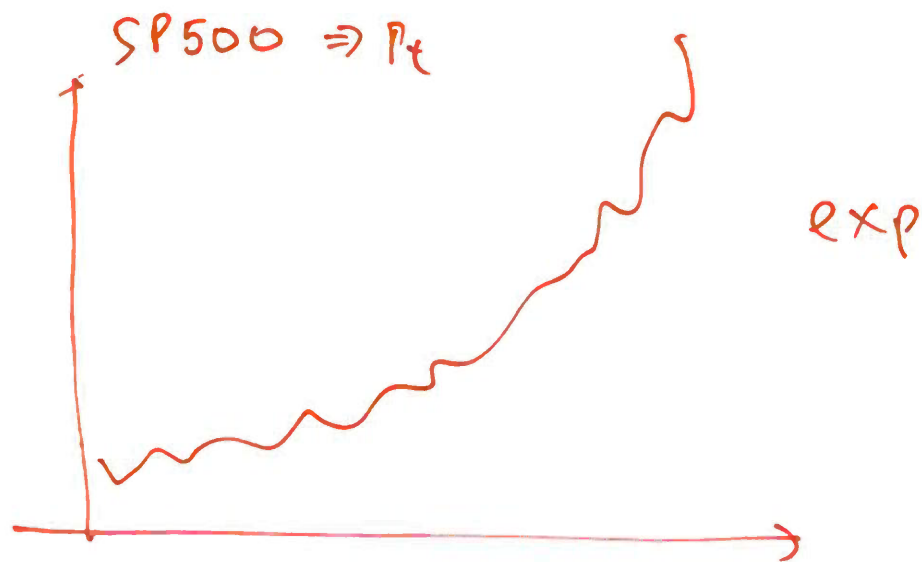
$$E(Y_t) = E(Y_{t-1}) = \dots = E(Y_{t-100}) = \mu$$

2. $\text{var}(Y_t) = \sigma^2$ does not depend on t

$$\text{var}(Y_t) = \text{var}(Y_{t-1}) = \dots = \text{var}(Y_{t-100}) = \sigma^2$$

3. $\text{cov}(Y_t, Y_{t-j}) = \gamma_j$ depends only on j but not on t , for $j = 1, 2, \dots$

Random Walk - Non-stationary



Fama: $\ln(P_t)$ is a R.W.

Random Walk Model

1. A useful model of **log** stock prices, p_t , is the **random walk model**.
2. Let ϵ_t be white noise with mean 0 and variance σ^2 .
3. Let p_0 be an arbitrary starting point.

Random Walk Model

The variable p_t is called a **random walk** with increments ϵ_t if:

$$\begin{aligned} p_t &= p_{t-1} + \epsilon_t, \quad t \geq 1 \\ &= p_0 + \epsilon_t + \dots + \epsilon_1. \end{aligned}$$

$$\epsilon_t \sim \text{WN}(0, \sigma^2)$$

$$E(\epsilon_t) = 0$$

$$\text{var}(\epsilon_t) = \sigma^2$$

$$t=1$$

$$p_1 = p_0 + \epsilon_1 \Rightarrow \epsilon_1$$

$$t=2$$

$$p_2 = p_1 + \epsilon_2$$

$$= p_0 + \epsilon_1 + \epsilon_2 \Rightarrow \epsilon_1 \text{ and } \epsilon_2$$

$$t=3$$

$$p_3 = p_2 + \epsilon_3$$

$$= p_0 + \epsilon_1 + \epsilon_2 + \epsilon_3$$

$$\epsilon_1, \epsilon_2, \epsilon_3$$

"3"

Assuming p_0 is fixed and for simplicity $p_0 = 0$. ←

1. The mean is

$$\mathbb{E}[p_t] = 0$$

because

2 eqn slide 9

$$\mathbb{E}(p_t) = \mathbb{E}(\epsilon_t + \dots + \epsilon_1) = \mathbb{E}(\epsilon_t) + \dots + \mathbb{E}(\epsilon_1) = 0$$

2. The variance is

$$\mathbb{V}[p_t] = \sigma^2 \cdot t \Rightarrow p_t \text{ is N-S}$$

because

$$\begin{aligned} \mathbb{V}(p_t) &= \mathbb{V}(\epsilon_t + \dots + \epsilon_1) = \mathbb{V}(\epsilon_t) + \dots + \mathbb{V}(\epsilon_1) + \text{cov}(\epsilon_t, \epsilon_{t-1}) \\ &= \sigma^2 + \dots + \sigma^2 \\ &= \sigma^2 t \end{aligned}$$

!!
0
WN

p_t is a R.W $\Rightarrow p_t$ is N-S

Random Walk Model with drift

Log stock prices, and more so market indices, tend to have a small and positive mean. Therefore, the random walk model becomes a random walk with drift:

$$\begin{aligned} p_t &= \mu + p_{t-1} + \epsilon_t, \quad t \geq 1 \\ &= \mu \cdot t + p_0 + \epsilon_t + \cdots + \epsilon_1. \end{aligned}$$

Random walk with drift

Assuming p_0 is fixed and for simplicity $p_0 = 0$.

The mean is

$$\mathbb{E}[p_t] = \mu \cdot t$$

N-S.

The variance is

$$\mathbb{V}[p_t] = \sigma^2 \cdot t$$

To test whether the log price of an asset follows a random walk or a random walk with drift, we employ the models:

①

$$p_t = \phi_1 p_{t-1} + \epsilon_t,$$

or

②

$$p_t = c + \phi_1 p_{t-1} + \epsilon_t.$$

This would require us to test:

$$H_0 : \phi_1 = 1 \Rightarrow p_t \text{ is R.W., } p_t \text{ is N-S}$$

$$H_1 : \phi_1 < 1 \Rightarrow p_t \text{ is Stationary, } p_t \text{ can be used for AR(1)}$$

cannot estimate
AR(1)

↑

Let $\Delta y_t = y_t - y_{t-1}$

↳ difference (diff in R)

More generally, for a given time series y_t , we can test for the existence of a unit root using the regression

$$y_t = c_t + \overset{\textcircled{2}}{\phi_1} y_{t-1} + \sum_{i=1}^{\overset{\textcircled{3}}{p}} \beta_i \Delta y_{t-i} + \epsilon_t$$

$- y_{t-1}$ $- y_{t-1}$ white noise

Here y_t is the series in levels.

The idea of the additional lagged differences $\Delta y_{t-1}, \dots, \Delta y_{t-p}$ is to model the autocorrelation so that the residuals of the equation are white noise. The best way to choose this lag length p is using BIC.

The hypotheses are

$$H_0 : \phi_1 = 1 \quad \text{versus} \quad H_1 : \phi_1 < 1.$$

$\underbrace{\phi_1 = 1}_{\text{RW unit root}} \quad \underbrace{\phi_1 < 1}_{\text{stationary}}$

A convenient way to test $H_0 : \phi_1 = 1$ is to subtract y_{t-1} from both sides, giving

slide 16

(2)

$$\Delta y_t = c_t + \phi_{c1} y_{t-1} + \sum_{i=1}^{p-1} \beta_i \Delta y_{t-i} + \epsilon_t,$$

slide 14

where

$$\phi_{c1} = \phi_1 - 1$$

$$\phi_1 y_{t-1} - y_{t-1}$$

$$(\phi_1 - 1) y_{t-1}$$

Here, Δy_t is the series in first differences.

Equivalently, the hypotheses can be written as

$$H_0 : \phi_{c1} = 0 \quad \text{versus} \quad H_1 : \phi_{c1} < 0.$$

$$\phi_1 = 1 \Rightarrow \text{R.W.}$$

stationary.

Thus, this is a left-tailed test.

Augmented Dickey-Fuller Unit Root Test

If y_t clearly does not have a time trend, then

$$c_t = c,$$

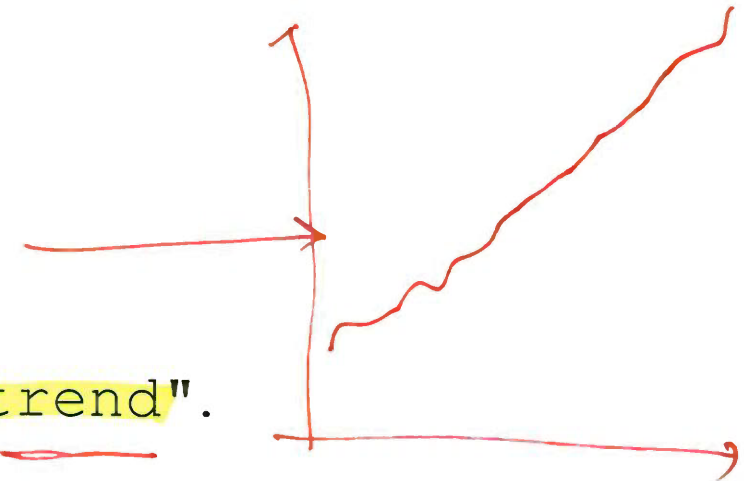
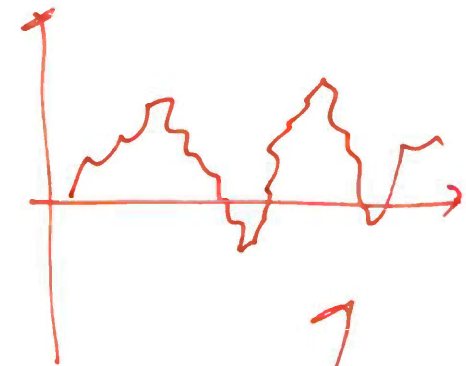
where c is a constant. In **R**, this is called `type = "drift"`.

OR

If y_t clearly has a linear time trend, then

$$c_t = c_0 + c_1 t$$

In **R**, this specification is called `type = "trend"`.



Augmented Dickey-Fuller Unit Root Test

The t-ratio

$$ADF = \frac{\hat{\phi}_{c1}}{\text{s.e.}(\hat{\phi}_{c1})},$$

slide 15

$$H_0 : \phi_c = 0$$

are referred to as the Augmented Dickey-Fuller (ADF) unit root tests.

The critical values for this statistic have been tabulated using simulations.

We reject the null hypothesis if the ADF test is less than the appropriate critical value.

left tail rejection.

not two-tail test.

ADF test in R

```
library(urca)
```

```
adf <- ur.df (y,  
              type = "trend", # "drift", or "trend"  
              selectlags = "BIC",  
              lags = 12, # max lag  
              )
```

→ slide 16 ⇒ see graph.

```
summary(adf)
```

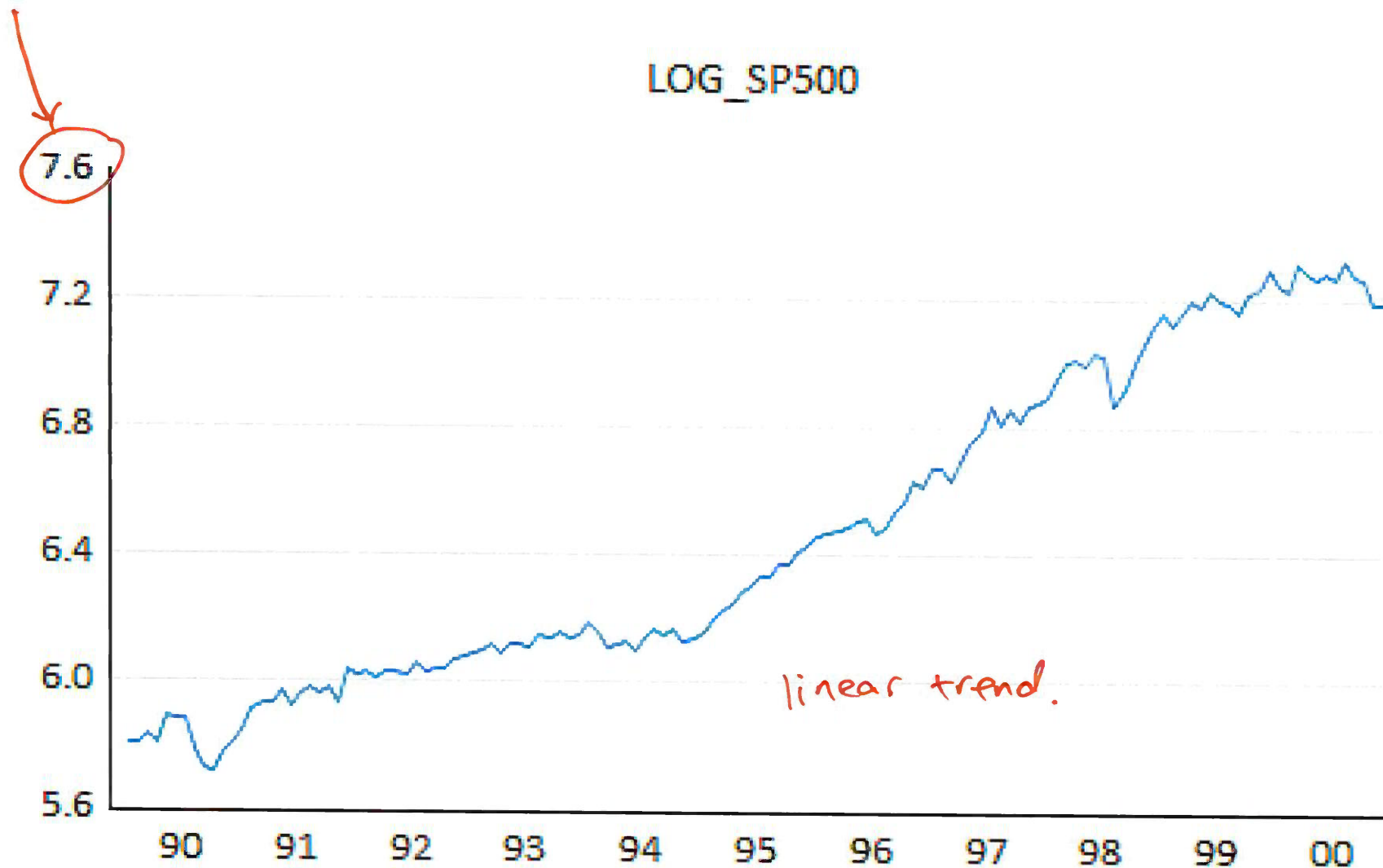
Example: Testing for a unit root in log stock prices

The log of asset prices is usually modelled as a random walk with drift.

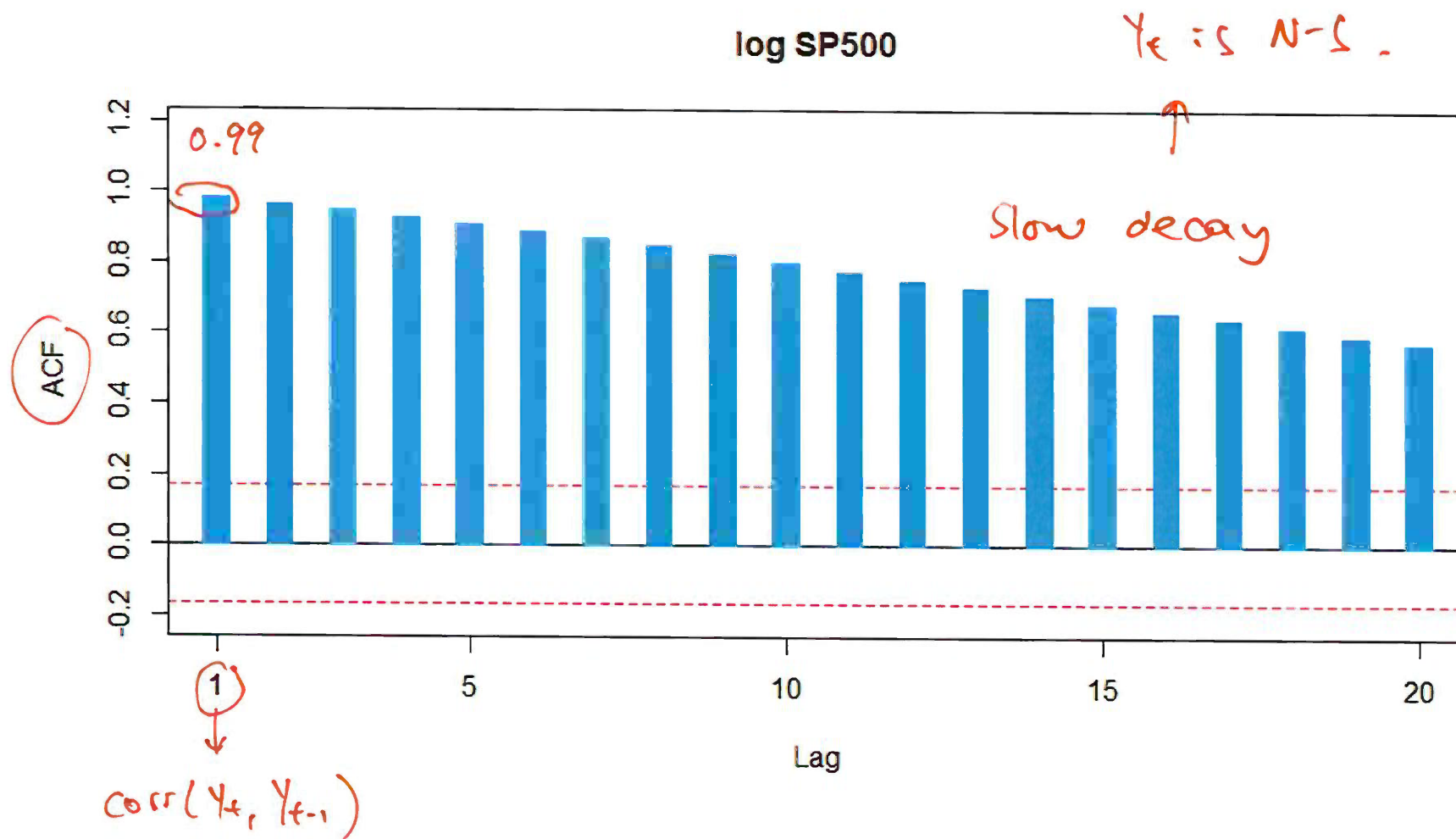
R.W

Consider testing for a unit root in the log of the monthly S&P 500 index, p_t over the period January 1990 through January 2001.

Log of S&P 500 index, Jan 1990 to Jan 2001



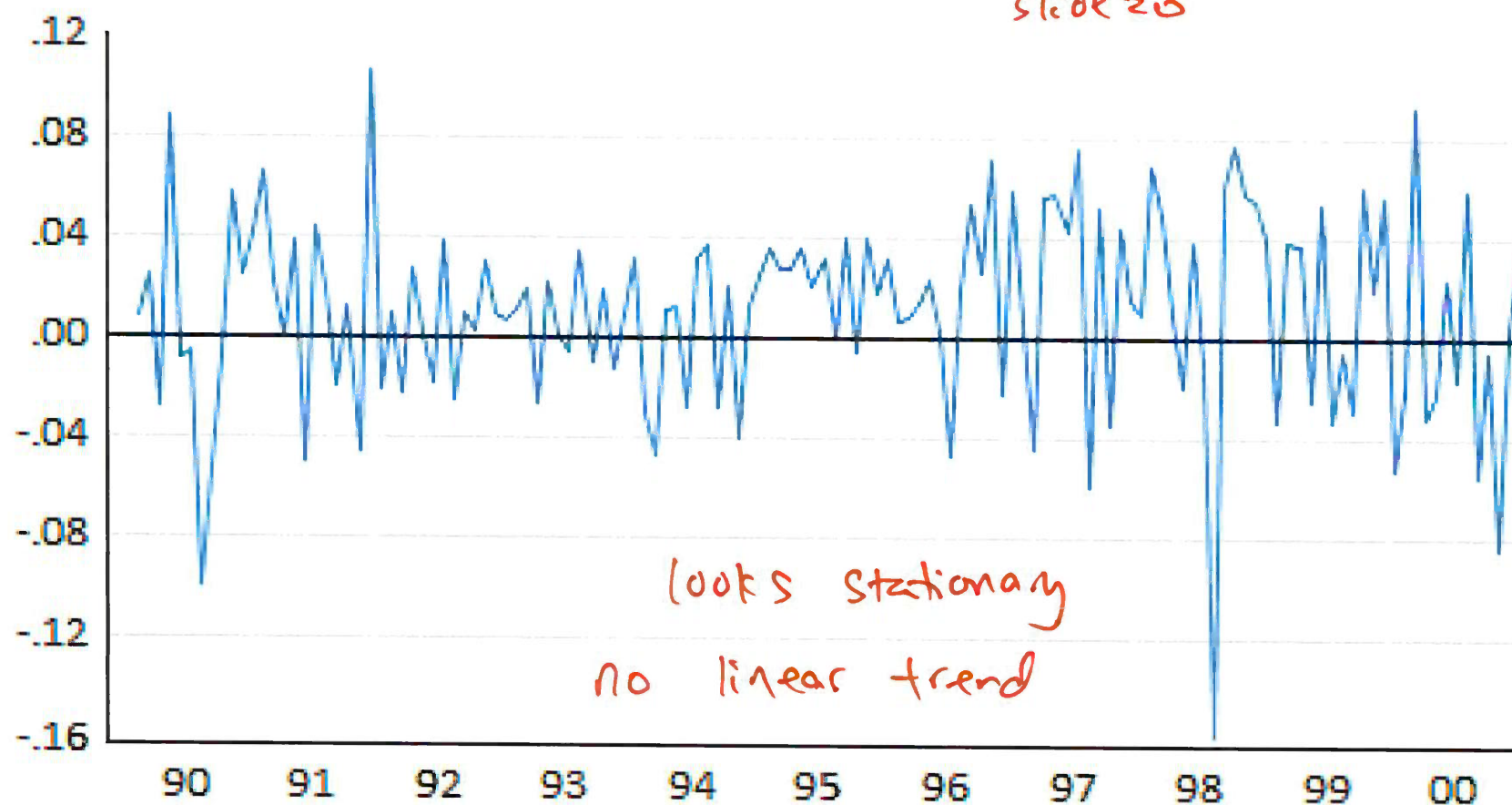
Q: is this R.W with drift?



Log Return of S&P 500 index

$$r_t = \log P_t - \log P_{t-1}$$

slide 20



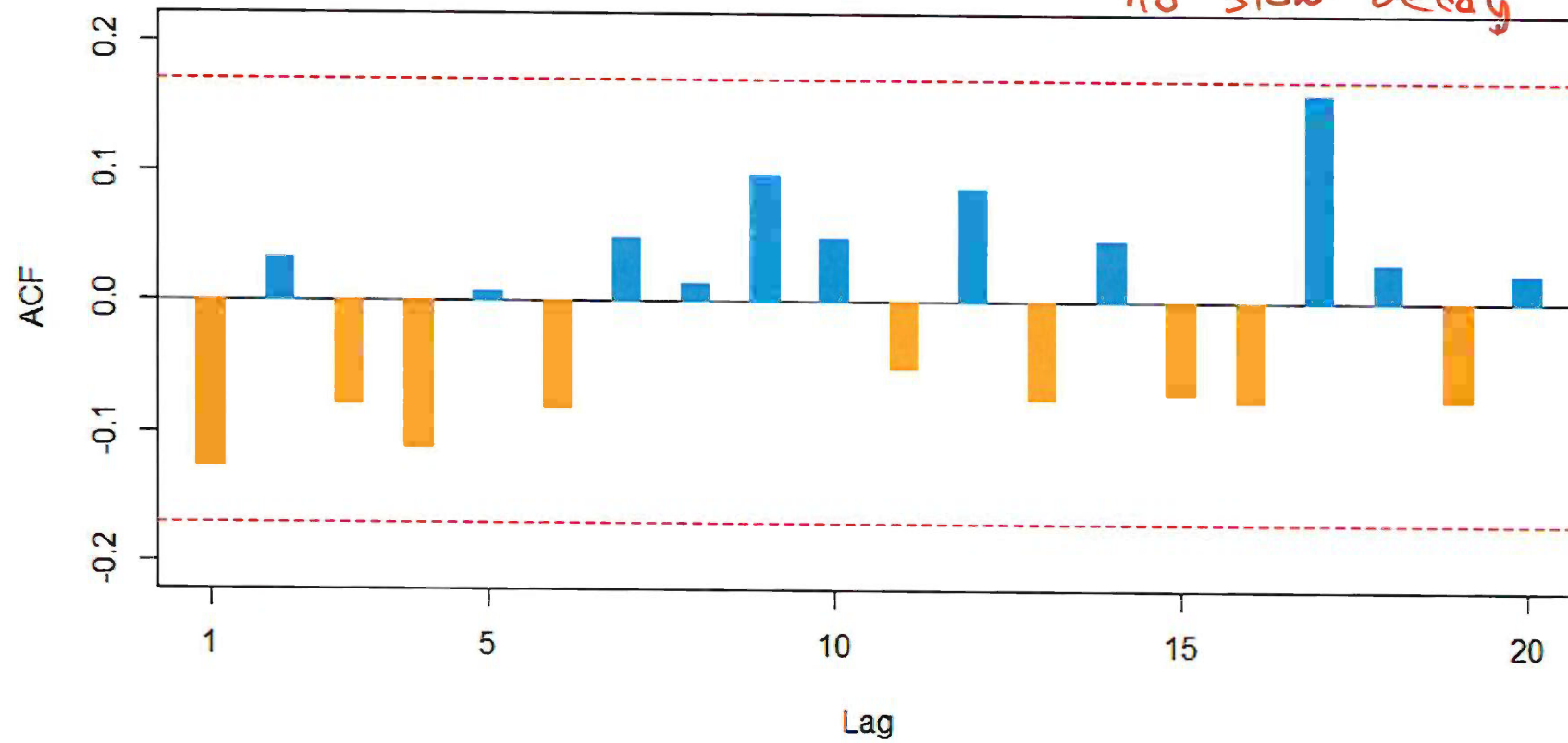
$$H_0: \phi_{c1} = 0$$

$$H_A: \phi_{c1} < 0$$

expect to reject H_0

Slide 31 $\Rightarrow$ reject H_0

log Return $\rightarrow$ seems stationary
no slow decay



Let us now conduct a statistical test to determine whether or not log of S&P 500 is non-stationary. This would require us to test, in the ADF regression:

$$H_0 : \phi_{c1} = 0$$

$$H_1 : \phi_{c1} < 0$$

The test statistic is

$$R \quad ADF = \frac{\hat{\phi}_{c1}}{\text{s.e.}(\hat{\phi}_{c1})} = -1.652 \quad \text{slide 26}$$

The 5% critical value is tau3 = -3.43. Since slide 27

$$-1.652 > -3.43$$

we fail to reject H_0 at the 5% significance level. Therefore, there is insufficient evidence to conclude that log(S&P 500) is stationary.

```
log_sp500 = log(data.xts$sp500)

adf_level <- ur.df(log_sp500,
                  type = "trend",
                  selectlags = "BIC",
                  lags = 12)

summary(adf_level)
```

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####
```

Test regression trend

call:

```
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.167406	-0.024106	0.000845	0.020012	0.089164

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.3191811	0.1850001	1.725	0.0871
z.lag.1	-0.0549851	0.0332845	-1.652	0.1012
tt	0.0007282	0.0004547	1.602	0.1120
z.diff.lag	-0.1474630	0.0919790	-1.603	0.1116

--- Δ $p \approx 1$
 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03794 on 116 degrees of freedom
 Multiple R-squared: 0.05354, Adjusted R-squared: 0.02907
 F-statistic: 2.187 on 3 and 116 DF, p-value: 0.09325

ADF = $\frac{(1)}{(2)}$

incorrect.

C_0
 y_{t-1}
 t
 Δy_{t-1}
 $\downarrow$
 slide 15
 16

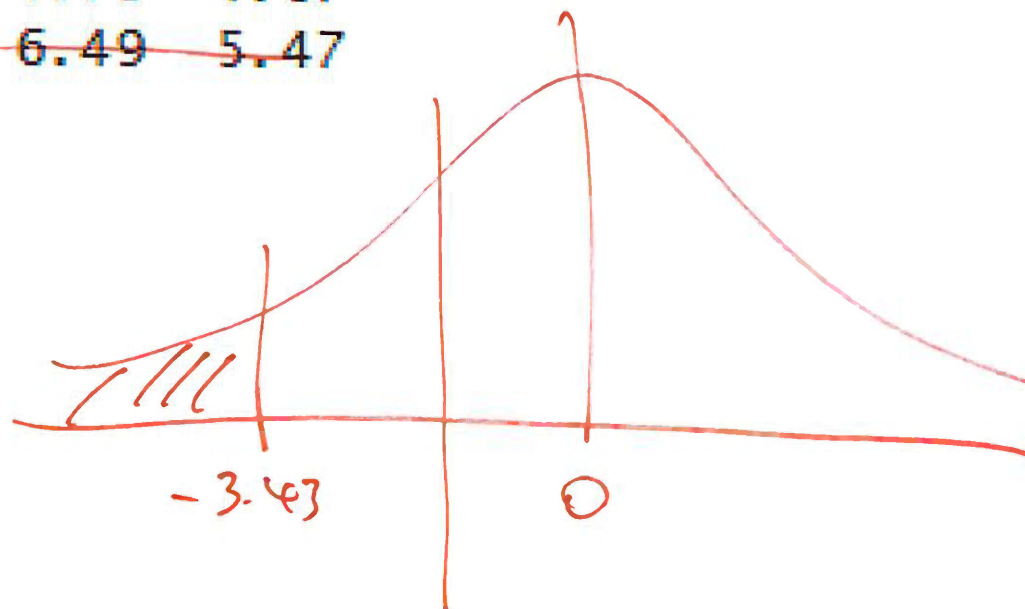
ϕ_{c1} (1) (2)

ADF

value of test-statistic is: -1.652 5.5772 1.3655

critical values for test statistics:

	1pct	5pct	10pct
→ tau3	-3.99	-3.43	-3.13
phi2	6.22	4.75	4.07
phi3	8.43	6.49	5.47



Fama is correct.



$\ln(\text{SP500})$



drift.
R.W.

-1.652 fail to reject $H_0: \phi_{c1} = 0$

Let us now conduct a statistical test to determine whether or not the log returns of S&P 500 is non-stationary. Again, in the ADF regression, we test:

$$H_0 : \phi_{c1} = 0$$

$$H_1 : \phi_{c1} < 0$$

The test statistic is

$$ADF = \frac{\hat{\phi}_{c1}}{\text{s.e.}(\hat{\phi}_{c1})} = -8.600 \quad \text{slide 30}$$

slide 31

The 5% critical value is $\tau_2 = -2.88$. Since

$$-8.600 < -2.88,$$

we reject H_0 at the 5% significance level and conclude that the log returns are stationary.

```
log_ret <- na.omit(diff(log_sp500))
```

```
# we select "drift"
```

```
adf_log.ret <- ur.df(log_ret,
```

type = "drift", no linear trend.

```
selectlags = "BIC",
```

```
lags = 12)
```

```
summary(adf_log.ret)
```

slide 22


```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####
```

Test regression drift

call:

```
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.17242	-0.02565	0.00233	0.02269	0.08406

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.013446	0.003838	3.503	0.000654	***
<u>z.lag.1</u>	-1.211969	0.140930	-8.600	4.41e-14	***
<u>z.diff.lag</u>	0.021691	0.091784	0.236	0.813594	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03798 on 116 degrees of freedom
 Multiple R-squared: 0.5968, Adjusted R-squared: 0.5899
 F-statistic: 85.86 on 2 and 116 DF, p-value: < 2.2e-16

ADF test

ϕ_{c1}

C
 y_{t-1}
 Δy_{t-1}
 ↓
 slide 15
 16

Value of test-statistic is: -8.5998 36.9863

Critical values for test statistics:

	1pct	5pct	10pct
tau2	-3.46	-2.88	-2.57
phi1	6.52	4.63	3.81

1:02

1:07 back.

Stationary Processes

not for fun.

AR(1) processes

Recall that we define $\mathcal{F}_{t-1}$ to be all information available to the econometrician at time $t - 1$.

This could consist of possible lagged values of y_t , such as $y_{t-1}, y_{t-2}, \dots, y_1$.

$$\mathcal{F}_{t-1} = (y_{t-1}, \dots, y_1) \quad \text{all info: up until "t-1"}$$

idea: predict y_t conditional $\mathcal{F}_{t-1}$

Definition

A time series ϵ_t is a martingale difference sequence (MDS) if

$$E(\epsilon_t \mid \mathcal{F}_{t-1}) = 0$$

$\hookrightarrow$ conditional on $\mathcal{F}_{t-1}$

Here $\mathcal{F}_{t-1}$ = the information set available up to time $t - 1$

$E(\epsilon_t \mid \mathcal{F}_{t-1})$ = the conditional expectation of u_t given past information.

shocks

↑

$E(\epsilon_t \mid \mathcal{F}_{t-1}) = 0$ means that the expected value of the current shock, given all past information, is zero.

AR(1) processes

A time series y_t is an **AR(1) process** if

$$y_t = c + \phi_1 y_{t-1} + \varepsilon_t$$

Assume ε_t is an MDS and $\mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) = \mathbb{V}(\varepsilon_t) = \sigma_\varepsilon^2 = \text{a number}$

$$\downarrow$$
$$\mathbb{E}(\varepsilon_t | \mathcal{F}_{t-1}) = 0$$

$\downarrow$ conditional $\uparrow \Rightarrow \mathcal{F}_{t-1}$ is useless.

$$-1 < \phi_1 < 1$$

ϕ_1 closer to 1 then $\forall t$ remember a lot.
 ϕ_1 closer to 0 then $\forall t$ forgets.

Unconditional Mean and Variance

We will show that: (a) the unconditional mean

$$\mathbb{E}(y_t) = \frac{c}{1 - \phi_1}$$

$$\phi_1 = 1$$

$E(y_t)$ is not define

(b) the unconditional variance

$$\mathbb{V}(y_t) = \frac{\sigma_\varepsilon^2}{1 - \phi_1^2}$$

$$\phi_1 = 1.1$$

$\text{var}(y_t) < 0$ not meaningful

An AR(1) process is **covariance (weakly) stationary** if $|\phi_1| < 1$.

AR(1) processes

$$\mathcal{F}_{t-1} \Rightarrow y_{t-1}, y_{t-2}, \dots$$

The **conditional mean** of y_t is

$$y_t = c + \phi_1 y_{t-1} + \varepsilon_t$$

best guess



$$\mathbb{E}[y_t | \mathcal{F}_{t-1}] = \mathbb{E}[c + \phi_1 y_{t-1} + \varepsilon_t | \mathcal{F}_{t-1}]$$

today

$$= \mathbb{E}[c | \mathcal{F}_{t-1}] + \mathbb{E}[\phi_1 y_{t-1} | \mathcal{F}_{t-1}] + \underbrace{\mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}]}_{0 \text{ assume MDS}}$$

$$= c + \phi_1 \mathbb{E}[y_{t-1} | \mathcal{F}_{t-1}] + 0$$

$$= c + \phi_1 y_{t-1}$$

yesterday.

hope ϕ_1 is close to one.

AR(1) processes

- ▶ We can estimate the AR(1) model using OLS:

- ▶ Choose $(\hat{c}, \hat{\phi}_1)$ such that

$$(\hat{c}, \hat{\phi}_1) = \arg \min \sum_{t=2}^T (y_t - c - \phi_1 y_{t-1})^2.$$

- ▶ This is a regression of y_{t+1} on a constant c and the regressor y_t .
- ▶ All OLS diagnostics apply.

$$E(y_t) = E\left(E(y_t | F_{t-1})\right) \quad \text{LIE}$$

Question:

What is the unconditional mean of y_t ?

What is $E(y_t)$?

Hint: Use LIE

use this
 $E(y_t | F_{t-1})$ to get $E(y_t) \Rightarrow$ use LIE
 conditional unconditional.
 week 3.

Unconditional Mean

$$\mathbb{E}(y_t) = \mathbb{E}(\mathbb{E}(y_t | \mathcal{F}_{t-1})) \quad \text{LIE}$$

↓ Slide 37

$$= \mathbb{E}(c + \phi_1 y_{t-1})$$

$$= c + \phi_1 \mathbb{E}(y_{t-1})$$

$$= c + \phi_1 \mathbb{E}(y_t) \quad \text{Stationary, } \mathbb{E}(y_t) = \mathbb{E}(y_{t-1})$$

$$\mathbb{E}(y_t) = \frac{c}{1 - \phi_1} \quad \text{Solve for } \mathbb{E}(y_t)$$

$$c = 0 \quad \mathbb{E}(y_t) = 0$$

Conditional variance of AR(1) processes

The **conditional variance** of y_t is

$$y_t = c + \phi_1 y_{t-1} + \varepsilon_t$$

$$\mathbb{V}(y_t | \mathcal{F}_{t-1}) = \mathbb{V}(c + \phi_1 y_{t-1} + \varepsilon_t | \mathcal{F}_{t-1})$$

$$= \mathbb{V}(c + \phi_1 y_{t-1} | \mathcal{F}_{t-1}) + \mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1})$$

$$= 0 + \mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1})$$

$$= \mathbb{V}(\varepsilon_t)$$

$$= \sigma_\varepsilon^2$$

slide 35

a number

$$v(y_t) = v\left(E(y_t | F_{t-1})\right) + E\left(v(y_t | F_{t-1})\right) \quad \text{LTV}$$

Question:

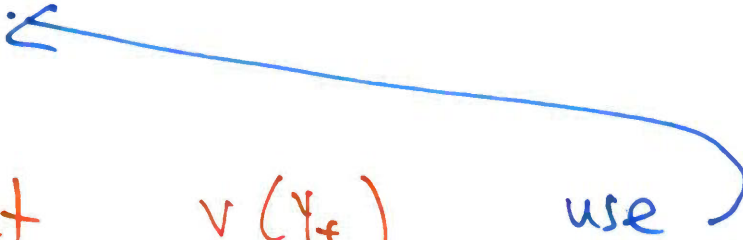
What is the unconditional variance of y_t ?

What is $\mathbb{V}(y_t)$?

Hint: Use Law of Total Variance

use this

~~var~~ $v(y_t | F_{t-1})$ to get $v(y_t)$ use
 conditional unconditional



Unconditional Variance

$$\mathbb{V}(y_t) = \mathbb{E}(\mathbb{V}(y_t | \mathcal{F}_{t-1})) + \mathbb{V}(\mathbb{E}(y_t | \mathcal{F}_{t-1})) \quad \text{LTV}$$

slide 41

slide 32

$$= \mathbb{E}(\sigma_\varepsilon^2) + \mathbb{V}(c + \phi_1 y_{t-1})$$

$$= \sigma_\varepsilon^2 + \phi_1^2 \mathbb{V}(y_{t-1})$$

$$= \sigma_\varepsilon^2 + \phi_1^2 \mathbb{V}(y_t)$$

$\text{Var}(y_{t-1}) = \text{Var}(y_t) \Rightarrow \text{Stationary}$

$$\mathbb{V}(y_t) = \frac{\sigma_\varepsilon^2}{1 - \phi_1^2}$$

= a number

~~Solve~~ Solve $v(y_t)$

A question

$$\text{MDS} \Rightarrow E(\varepsilon_t | \mathcal{F}_{t-1}) = 0$$

↑ ↑
gap = 1 period

What is

$$E(\varepsilon_{t+2} | \mathcal{F}_t) = 0$$

↑ ↑
gap $\Rightarrow$ 2 ~~period~~ period $\Rightarrow$ NOT
MDS

Answer: Apply LIE

$$E(\varepsilon_{t+2} | \mathcal{F}_t) = E(E(\varepsilon_{t+2} | \mathcal{F}_{t+1}) | \mathcal{F}_t) = E(0 | \mathcal{F}_t) = 0.$$

↑ ↑
gap $\Rightarrow$ 1 period

$$E(\varepsilon_{t+2} | \mathcal{F}_{t+1}) = 0 \text{ MDS}$$

Using LIE

In general, for $h \geq 1$ we have

$$E(\varepsilon_{t+h} | \mathcal{F}_t) = 0$$

because LIE

$$E(\varepsilon_{t+h} | \mathcal{F}_t) = E(E(\varepsilon_{t+h} | \mathcal{F}_{t+h-1}) | \mathcal{F}_t)$$

gap = 1-period $\Rightarrow$

$$= E(0 | \mathcal{F}_t)$$

$$E(\varepsilon_{t+h} | \mathcal{F}_{t+h-1}) = 0$$

Just
minus one

$$= 0$$

One-step ahead forecast

$$y_{t+1} = c + \phi_1 y_t + \varepsilon_{t+1} \quad \text{AR(1)}$$

replace "t" with "t+1"

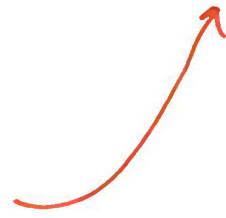
A one-step ahead forecast

best guess



$$\begin{aligned} \mathbb{E}(y_{t+1} | \mathcal{F}_t) &= c + \phi_1 \mathbb{E}(y_t | \mathcal{F}_t) + \overbrace{\mathbb{E}(\varepsilon_{t+1} | \mathcal{F}_t)}^{\text{MDS}} \\ &= c + \phi_1 y_t + \underbrace{0}_{\substack{\text{gap} = 1 \text{ period} \\ \parallel \\ 0}} \end{aligned}$$

$$\mathbb{E}(y_t | \mathcal{F}_t) = y_t$$



hope that ϕ_1 is close to 1

What about more than one-step ahead?

A **two-step ahead** forecast would be based on

best guess



$$\mathbb{E}(y_{t+2}|\mathcal{F}_t) = \mathbb{E}[c + \phi_1 y_{t+1} + \varepsilon_{t+2}|\mathcal{F}_t]$$

replace "t" with "t+2"

not
t+1

$$= c + \phi_1 \mathbb{E}[y_{t+1}|\mathcal{F}_t] + \mathbb{E}[\varepsilon_{t+2}|\mathcal{F}_t]$$

time forecast not actual value.

slide 46

$$= c + \phi_1 (c + \phi_1 y_t) + 0$$

slide 44

$$= c(1 + \phi_1) + \phi_1^2 y_t \quad \leftarrow$$

What about more than one-step ahead?

A three-step ahead forecast

replace "t" with "t+3" AR(1)

$$\begin{aligned}\mathbb{E}(y_{t+3}|\mathcal{F}_t) &= c + \phi_1 \mathbb{E}(y_{t+2}|\mathcal{F}_t) + \mathbb{E}(\varepsilon_{t+3}|\mathcal{F}_t) \\ &= c(1 + \phi_1 + \phi_1^2) + \phi_1^3 y_t.\end{aligned}$$

slide 47 ↓ slide 45
 0

etc...

Comments on the use of AR models for forecasting

- ▶ Forecasts based on well-specified AR models are consistent (i.e. bias is negligible in large samples).
- ▶ But we need to choose our lag length p carefully!
- ▶ Include enough lags to ensure that there is **no left over serial correlation in the residuals**.
- ▶ Then we can interpret ε_t as ‘news’ (unpredictable market forces).